LESSON PLAN

A/C MATERIAL (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	16	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Overall consideration of aircraft loading action.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Overall consideration of aircraft loading action. Symmetric flight loads, symmetric maneuvers
5. Specific Objectives	15.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of structural element diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary structural elements of an aircraft?
(b) Define symmetric flight in the context of loading.
(c) List three types of symmetric maneuvers.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Aircraft structural elements overview • Wing and fuselage load paths • Symmetric flight load characteristics • Maneuver load factors definition • Structural response to symmetric loads	L & Ds	PPT, IB	15 min
(b)	• Analysis of symmetric maneuver loads • Distribution of loads in symmetric flight • Effect of load factors on structure • Comparison of steady vs maneuver loads • Identification of critical structural points	L & Ds	PPT, PDF	20 min
(c)	• Summary of symmetric loading principles • Key structural element functions • Safety margins in symmetric loads • Preparation for aerodynamic distribution • Q&A on structural elements	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how symmetric maneuvers affect aircraft structure.
(b) Differentiate between symmetric flight loads and maneuver loads.
(c) Identify the main structural elements involved in symmetric loading.

12. Home Assignment and Follow Up	Prepare a brief note on how the structural elements discussed in this lesson will be analyzed in LP 02 regarding aerodynamic loading distribution.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Aerodynamic loading distribution, span wise and chord wise distribution, effects of sweep and distribution.	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Commander, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min